is another important step to encourage private players to show interest and look at this opportunity in a positive way. This, coupled with substantial investment from industry players, would result in a joint effort by both parties involved, which is necessary to build the ecosystem in the country.

To further tackle this issue, Chhabra suggests that technology transfer could be another way to fuel growth in the sector, similar to how the automotive industry in the country grew, which was primarily through collaboration with foreign OEMs. He says, "Collaborations for technology transfer are really a foot in the door because, if we can get partners, I think that would take us a lot of steps forward."

Local infrastructure

Interestingly, while many are still fazed by the lack of a component ecosystem to build IoT devices, most tend to forget one crucial aspect—cloud infrastructure. Given the amount of dependency domestic IoT players and global companies operating in the country have on AWS, Google Cloud, Microsoft Azure and its likes, especially, building reliable local cloud infrastructure is one of the cornerstones to make India reliable in terms of its IoT needs.

In fact, the indispensability of these giant cloud providers leaves very little space for domestic players who are providing cloud infrastructure, to which Chhabra points out, "If you really look at it, then their (domestic players') penetration would be less than 2% in the market because of the 98% being taken away by these global players."

A large part of this is because of the service-level agreements (SLAs) provided by global players, which are more competitive than the ones provided by the domestic. However, the latter argue that the demand for their services is much less, making it difficult for them to invest in SLAs to provide as many facilities as those offered by Google or Amazon.

Global giants have marked territories with their products that promise more reliability and have made themselves an inseparable part of the ecosystem, which is why it is obvious for a lot of companies working in the IoT space who choose to lean towards them.

While all of this seems to work well, data localisation still stands to be a significant challenge in using the services of global cloud providers. At this stage, data localisation is seen with respect to financial services only, though IoT as well is very reliant on it. For example, there is potential of a serious privacy breach when someone uses a device made in China, which is where all their data goes and is stored. Hence, data localisation particularly becomes THE most important thing to be addressed, especially when considering a local cloud infrastructure.

The only way to break this loop, as per Chhabra, is to make domestic cloud infrastructure come close to the reliability and accessibility of the global cloud, through government policies that promote the growth and use of indigenous cloud infrastructure and create a level playing field for them. He adds, "If we (as a country) can really create that demand then I'm sure that these companies would really come in and come up with better SLAs. This starts with businesses supporting and hiring local cloud infrastructure players."

Missing pieces

Coming to another crucial part of an IoT product, connectivity modules that are made in India are yet to garner mass usage in the country, even from domestic IoT players. The biggest challenge that we face is that countries around the globe, particularly in China, have reached economies of scale which we, unfortunately, haven't. The good part is that we can start with manufacturing connectivity modules in India, which would further be helped if businesses using these modules begin to use only India-made modules.

Again, rationalising custom duty is another great move that the government can make to encourage local businesses to opt for locally-made products. If the companies buying these modules give an assurance that they would buy locally then, obviously, we do not have to look for imports outside India. Chhabra adds, "From another standpoint, there is a need to ramp up investments on test facilities for RF, etc, so that products can be created which are also FDC approved, suggesting we can look for the export market for these as well."

Especially with the advent of 5G in the commercial broadband space right now, creating compatible products is needed. However, when experimenting with such a new-age technology, overcoming development hindrances to ensure that the technical aspects are right to make the product succeed is crucial. For this, the presence of sufficient testing facilities, particularly for start-ups and SMEs, is necessary.

The road to make India self-reliant is a huge opportunity and a challenge at the same time. Building a local ecosystem for IoT players to grow and scale up is definitely important, but also the requirement of domestic players to support this growth goes hand in hand. There is a need to add value to each product that is created, which can only be achieved when new and existing companies make a conscious effort to find the missing pieces and solve the puzzle. If this is kept as the supreme aim, India would be self-reliant in no time. EFY

The article is based on presentations 'How India is Going to Make Itself Self-Reliant wrt Growing IoT Demand?' by Arpit Chhabra and 'Making IoT Start Ups Successful' by Sanjeev Keskar made during Tech World Congress 2020. It has been prepared by Siddha Dhar, a business journalist at EFY.